Response to Referee #4

We thank the referee for a very constructive and thoughtful report. The comments help clarify the scope of the paper and point to extensions that strengthen the presentation of our results. Below we provide detailed, point-by-point responses. We quote the referee's comments in italics.

1. Sovereign default risk

Referee comment:

> *Ecuador defaulted in 2008 and 2020, yet the model does not include any features that capture sovereign default risk. One tractable way of including default risk is that of Corsetti et al. (2013) and Cantore et al. (2019).*

Response:

Thank you for raising this important modeling dimension. We agree that Ecuador's credit history raises the question of explicit sovereign default. Our model uses an endogenous risk-premium mechanism that links spreads to the stock of external debt, a conventional approach in small open economy DSGE models that allows us to match the large spread fluctuations observed in the data. Because the estimation window (2004-2019) ends before the 2020 default, the 2008 restructuring episode is absorbed through large estimated spread shocks, which the model interprets as sudden changes in market pricing of default risk.

To address this comment, we now: (i) explicitly acknowledge the absence of discrete default events in the model; (ii) discuss the frameworks of Corsetti et al. (2013) and Cantore et al. (2019) and explain their advantages as tractable ways to map fiscal fundamentals into sovereign spreads; and (iii) clarify that incorporating a full sovereign-default block would require solving an occasionally-binding problem with an endogenous default decision, restructuring terms, and multiple equilibrium values—an extension that is conceptually valuable but significantly beyond the scope of the present paper. We also note that our main results, which concern the differential performance of a fiscal consolidation plan under alternative commodity-revenue paths, operate through the debt dynamics and risk-premium mechanism rather than the default decision per se. See p. 8, 2nd paragraph, and footnote 10.

2. Symmetry of responses and the suggestion to use a nonlinear perfect-foresight solution

Referee comment:

> *As shown in Figure 4, the effects of higher or lower oil prices are symmetric. This is not surprising since the model is solved at first order. One potential solution would be to solve the model under perfect foresight to preserve nonlinearities.*

Response:

We appreciate this suggestion. It is correct that a first-order stochastic solution exhibits certainty equivalence and therefore rules out true state-dependence. At the same time, it is worth noting that in our setting the fiscal outcomes under higher versus lower oil revenues are already not fully symmetric even at first order. This is because the two scenarios map into different levels of the endogenous state variables, in particular the debt-to-GDP ratio, and the risk-premium schedule is a nonlinear function of that ratio. As a result, higher-revenue paths push the economy onto lower-debt regions of the risk-premium curve, while lower-revenue paths place the economy in higher-debt regions with steeper premia. These differences generate asymmetric responses despite the linearization of the rest of the model.

Nonetheless, we agree entirely that simulating the consolidation program under perfect foresight allows us to retain all nonlinearities of the structural model and to include anticipation effects, as you suggest in the next comment.

In the revised paper, we therefore simulate the full nonlinear model under perfect foresight. Specifically:

- We take the posterior mean of the structural parameters estimated from the linearized stochastic model.
- We load these parameters into the exact nonlinear system presented in the theoretical section.
- We implement the IMF-program paths for fiscal instruments $(g_t, i_t^g, tr_t, \tau_t^c, \tau_t^{lab}, \tau_t^k)$ as *exogenous deterministic sequences*. In practice, we declare them as `varexo_det` objects in Dynare and feed their quarter-by-quarter values as specified by the program.
- We specify the deterministic paths for oil revenue for (i) the baseline program, (ii) a one-standard-deviation higher path, and (iii) a lower path.
- We solve the full transition using `perfect_foresight_setup` and `perfect_foresight_solver`.

This procedure delivers transition dynamics with full anticipation and preserves all nonlinear interactions in the model. The results confirm—and reinforce—the qualitative asymmetry documented in the stochastic simulations: a low-revenue scenario leads to insufficient fiscal adjustment and rising spreads, whereas a high-revenue scenario produces substantial over-performance of the fiscal targets and declining spreads. The implementation of the new perfect-foresight simulation is documented in Section 6 (p. 21, 1st paragraph).

3. Anticipation of the fiscal program and credibility

Referee comment:

> *The approach used in the paper does not account for the fact that agents know the IMF program. The model should be solved under perfect foresight so that expectations respond as soon as the new policy is known. The paper should also discuss the credibility of the fiscal plan.*

Response:

The referee correctly notes that the estimation-based conditional forecast uses a sequence of unexpected shocks to reproduce the fiscal paths, which abstracts from anticipation effects. We agree that a perfect-foresight solution would clarify the consequences of full program credibility.

In the revised manuscript, the results follow a deterministic simulation of the nonlinear model in which the entire IMF fiscal program is announced at time 0 and thus fully anticipated. As described above, this is implemented by treating each fiscal instrument as an exogenous deterministic variable with a known future path. This approach is computationally tractable, avoids the need to introduce multi-period news shocks for multiple fiscal instruments, and produces the correct expectation formation under full credibility. The implementation of the new perfect-foresight simulation is documented in Section 6 (p. 21, 1st paragraph).

The perfect-foresight simulations show that incorporating anticipation effects does not overturn our main findings. Anticipation changes the timing of some private-sector adjustments (e.g., consumption and investment), but the qualitative contrast across revenue scenarios remains: under low oil revenues, debt and spreads remain high along the program path, whereas under high oil revenues the consolidation over-performs and financing conditions improve.

We additionally expand the discussion of program credibility. Ecuador's historical fiscal slippages and episodes of policy reversal mean that perfect credibility is an unrealistic assumption. We therefore note explicitly that limited credibility would *amplify* the risks associated with low oil revenue paths: if agents expected possible future deviations from the announced fiscal plan, the risk premium would incorporate an additional credibility spread, worsening the macro-fiscal dynamics. A full formal treatment of imperfect credibility would require a Markov-switching or occasionally-binding policy regime, which is beyond the scope of this paper but is now clearly identified as an important extension for future work. See p. 27, 4th paragraph, and footnote 35.

4. IMF financing flows

Referee comment:

> *A key aspect of the analysis is that the fiscal consolidation is based on an Extended Fund Facility provided by the IMF, which involves financial assistance. Yet external transfers are not present in the model.*

Response:

We thank the referee for raising this point. The IMF's Extended Fund Facility provides official loans at concessional rates rather than transfers. Our model does not distinguish official from market borrowing; instead, it treats all external liabilities in reduced form through the country-risk-premium schedule. Importantly, the fiscal-consolidation simulations do not impose any particular composition of financing during the program; public debt evolves endogenously given the fiscal-instrument paths.

In principle, concessional official loans could reduce interest expenditures relative to market financing. However, this effect is quantitatively limited in our context: IMF disbursements represent a small share of public debt, and our consolidation experiments condition on the program's fiscal-policy paths rather than on the government's optimal financing mix. Moreover, the model's spread-debt mechanism already captures the primary stabilizing channel of official support—namely, that easier financing conditions reduce borrowing costs—without requiring explicit modeling of separate debt classes.

In the revised paper, we now clarify this interpretation and explain that introducing a distinct official-lending block (with concessional pricing or maturities) is feasible but would not materially alter the consolidation dynamics we study. See p. 12, 3rd paragraph, and footnote 29 on p. 22.

5. Transfer rule under a consolidation program

Referee comment:

> *Transfers are procyclical in the estimated rule. It is not clear that the government would maintain such a rule during a fiscal consolidation programme.*

Response:

This is a valid concern. Transfers appear procyclical in the estimated reaction function because the rule is meant to capture Ecuador's historical fiscal behavior over the estimation sample. However, during the fiscal consolidation simulations, the transfer path is not governed by this rule. Instead, the program's explicit transfer targets are imposed directly as deterministic sequences under the perfect-foresight procedure (previously implemented via conditional shocks). We now clarify this point in the text: the government's expenditure reaction functions, including for transfers, are used solely for estimating policy behavior in pre-consolidation data and do not constrain or determine the consolidation trajectory. See p. 21, 1st paragraph.

6. Prior distributions

Referee comment:

> *Prior distributions should not be based on the same sample used for estimation.*

Response:

We thank the referee for raising this important point regarding the construction of prior distributions.

We agree that, in principle, priors should reflect either independent information or beliefs not derived from the estimation sample. In our setting, however, there is no well-established empirical evidence on the behavior of fiscal expenditure rules for Ecuador—particularly regarding their cyclicality or responsiveness to public debt—which makes it difficult to discipline priors using external studies.

Using pre-sample data is also not feasible in our case, as consistent and reliable quarterly series for fiscal variables are not available prior to the estimation period. This limits the scope for constructing priors based on independent historical information.

Our approach should therefore be interpreted as a pragmatic empirical-Bayes strategy. Specifically, we use simple OLS regressions of the fiscal rules on output and debt gaps—estimated outside the structural model—to obtain reasonable central values for the prior means. These auxiliary estimates are not intended to provide identification, but rather to anchor the priors in empirically plausible regions of the parameter space and avoid economically implausible configurations.

Importantly, the priors remain relatively diffuse, so that the posterior distributions are primarily shaped by the likelihood of the full DSGE model rather than by the prior means. In this sense, identification is driven by the Bayesian estimation of the structural model, not by the auxiliary regressions used to center the priors.

We have clarified this point in the revised manuscript by explicitly stating the role of these auxiliary regressions and the relatively loose nature of the priors. See Appendix D, p. 45, 2nd paragraph.

7. IRFs and uncertainty

Referee comment:

> *It is not clear whether the IRFs are based on the posterior mean or median. It might be desirable to plot the mean of the IRFs generated by drawing parameters from the posterior distribution.*

Response:

We appreciate this suggestion. All impulse responses reported in the main text are computed using the posterior mean of the estimated structural parameters, and we now clarify this explicitly. The IRFs to oil-price and risk-premium shocks are included to illustrate the mechanisms of the model, and for this purpose the posterior-mean representation is standard.

To address the referee's concern about parameter uncertainty in a way that is most informative for our main quantitative findings, we now complement the fiscal-consolidation experiment with a Bayesian assessment of uncertainty. Specifically, in Appendix J (p. 64), we report the transition paths of key macroeconomic variables under the baseline oil-revenue trajectory together with 90% credible intervals constructed from draws of the parameter vector from the posterior distribution (a subset of 200 draws). Each draw is fed into the nonlinear perfect-foresight simulation, generating a distribution of deterministic transition paths. We chose a subset of draws and not the full set because solving the model under perfect foresight takes a long time and our interest was to show the role of parameter uncertainty in the simulation.

This exercise provides a transparent measure of the model's uncertainty around the fiscal-consolidation dynamics, which is the central object of interest. In contrast, our auxiliary IRFs are intended primarily to illustrate the intuition for the transmission mechanisms, and we therefore retain them at the posterior-mean parameter values. We have revised the text to make these points clear.

Minor comments

- On estimation and simulation periods in the abstract: Thanks for this comment. We now say "The model is estimated using Ecuadorian data for 2004—2019 and then used to simulate Ecuador's 2020—2025 IMF-supported fiscal consolidation under perfect foresight, treating the program's fiscal instruments as fully anticipated policy paths."

- On optimizers and restricted agents: We thank the referee for this comment and agree that both household types optimize. We use the terms "Ricardian" and "hand-to-mouth" to emphasize differences in access to financial markets: Ricardian households solve an intertemporal problem with saving and borrowing, while hand-to-mouth households solve a static problem due to the absence of asset markets. We avoid the term "non-Ricardian" because it is often used more broadly in the literature and may be less precise in describing the constraint faced by these households. We have clarified this distinction in the text.

- On quarters in the sample: We now specify our sample as 2004:Q1 to 2019:Q4.

- On the use of the HP filter: We thank the referee for this comment. We recognize that this approach differs from standard DSGE estimation practices emphasized in the literature (e.g., Pfeifer), where filtering is typically aligned with the model's state-space representation. In our case, however, the HP filter is applied only to a small set of exogenous processes (oil production, oil prices, world GDP, and the international interest rate) to extract their stationary components, which are then modeled as AR(1) processes. These variables enter the model as exogenous inputs and are not part of the estimated state-space system. All endogenous observable variables are expressed in quarter-over-quarter growth rates (after deflation), ensuring consistency between the data and the model's stationarization. The only exception is hours worked, for which we apply the two-sided HP filter. Although the desirable strategy would have been to use a one-sided filter, we find that the difference between the one- and two-sided filtered series is very close to zero while the correlations of each series with non-oil GDP growth are -0.0251 for the two-sided and -0.0284 for the one-sided series. For these reasons, we do not believe the use of the HP filter significantly affects the internal consistency of the estimation or the interpretation of the results, in particular because hours worked is only one out of the nine endogenous observable variables.

Concluding remark

We are grateful for the referee's insightful comments. In response, the revised paper adopts a nonlinear perfect-foresight framework in which the fiscal measures in the IMF program are fully anticipated and implemented as exogenous deterministic paths, directly addressing symmetry and anticipation. The estimation–simulation connection is preserved by using the posterior-mean parameters and, in an appendix, we report credible bands based on posterior draws for the consolidation transition. We also clarify our modeling choices regarding sovereign risk, the treatment of IMF financing, the transfer rule under the program, and prior construction. These changes, together with the uncertainty assessment, materially enhance the paper's clarity and robustness.

Sincerely,
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References

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Corsetti, Giancarlo, Keith Kuester, André Meier, and Gernot J. Müller. (2013) "Sovereign Risk, Fiscal Policy, and Macroeconomic Stability." *The Economic Journal*, 123, F99–F132.